

EDUCATION

National University of Singapore

Ph.D. in Computer Science, advised by Prof. Tulika Mitra

Singapore

Aug 2026–Present

Peking University

B.S. in Applied Physics, Department of EECS, CGPA: 89/100 (WES-calculated: 3.85/4.00)

Beijing, China

Sep 2022–Jul 2026

RESEARCH EXPERIENCE

Agentic Foundation Model Training Framework

Research Assistant to: Yanbo Liang & Tingfeng Ruan

ByteDance Seed

Mar 2026–Present

- Building Hermes, an agentic system that optimizes foundation-model training with `torch.compile`.
- Developing an agent that automatically writes and applies FX graph passes for graph-level optimization.
- Developing an agent that authors Inductor IR passes to discover new fusion patterns.
- Designing an agent that generates joint FX graphs for scheduling large-scale distributed training workloads.

Stanford UGVR Program

Research Assistant to: Prof. Thierry Tamine

Stanford University

Jun 2025–Jan 2026

- Extended an ASIP design tool with memory-aware profiling and logging capabilities.
- Collected and analyzed memory-access traces across ASIP designs and workloads to evaluate architectural trade-offs.
- Implemented multicore simulation and developed a profiling-driven optimizer for memory-aware algorithms and compilation.
- Presented the work at [Synopsys ASIP University Day 2025](#).

Lossless Compression Accelerator Architecture Design

Research Assistant to: Prof. Bonan Yan

Peking University

Sep 2024–Jun 2026

- Designed an end-to-end workflow for a lossless-compression accelerator based on probabilistic circuits.
- Developed a bit-exact Python implementation of the rANS compression algorithm.
- Proposed a parallel accelerator architecture to improve computational efficiency.

Quantization Research and Deployment of LLMs

Research Assistant to: Senior Engineer Fan Wang

HOUMO.AI

Jun 2024–Aug 2024

- Evaluated multiple quantization algorithms for BERT- and Llama-family models on production hardware.

Research Rotations

Research Assistant to: Profs. Yuchao Yang, Bonan Yan, and Kechao Tang

Peking University

Feb 2024–May 2024

PUBLICATIONS

- [1] Anjunyi Fan, Xuejie Liu, Anji Liu, Qiuping Wu, Jiaqi Yang, **Yuchao Qin**, Guy Van den Broeck, Yitao Liang, and Bonan Yan. 2026. ESTroM: element-flow architecture for processing sparse tractable probabilistic models. In *Proceedings of the 32nd International Symposium on High-Performance Computer Architecture (HPCA)*. (Jan. 2026).

- [2] **Yuchao Qin**, Anjunyi Fan, and Bonan Yan. 2025. RAS: a bit-exact rANS accelerator for high-performance neural lossless compression. (2025). <https://arxiv.org/abs/2511.04684> arXiv: 2511.04684 [cs.AR].
- [3] **Yuchao Qin**, Peijing Li, and Thierry Tambe. 2025. ChronoMem: evaluating data lifetime and optimizing data placement for retention constrained on-chip memories in ASIPs. In *Synopsys ASIP University Day*. Presentation. Synopsys. (Nov. 2025). <https://www.synopsys.com/content/dam/synopsys/events/documents/asipunivday2025-yuchao-qin.pdf>.

COURSE PROJECTS

- **Five-Stage RISC-V CPU Core:** Built a five-stage RISC-V core with branch prediction; won first prize in the China College IC Competition. [Project repository](#).
- **CNN Accelerator Tape-Out:** Designing a CNN accelerator for tape-out in a TSMC 180-nm process. [Project repository](#).
- **Sparse Matrix–Dense Matrix Multiplication Accelerator:** Developed a hardware accelerator for SpMM using a high-level chip-design language. [Project repository](#).
- **FPGA Detailed Placement:** Developed an EDA tool for FPGA detailed placement and wirelength optimization. [Project repository](#).

SKILL SUMMARY

- **Languages:** Mandarin (native), English
- **Programming:** C++, Python, MATLAB, Julia, ASIP design languages
- **Hardware Design and Simulation:** Verilog, Chisel, Cadence Virtuoso, ASIP design tools

SCHOLARSHIPS AND AWARDS

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|---|-----------|
| • Silver Medal in the 38th Chinese Physics Olympiad Finals | 2021 |
| • Jiukun Scholarship | 2022–2023 |
| • Merit Student at Peking University | 2023 |
| • Most Valuable Player, Peking University Soccer Cup | 2023 |
| • TikTok Scholarship for Electrical Engineering Students | 2023–2024 |
| • Selected for the inaugural Experimental Class in Electronic Information Science | 2023–2026 |

LEADERSHIP AND ACTIVITIES

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| <ul style="list-style-type: none"> • Captain, Peking University EECS Soccer Team
<i>Led the team to three consecutive semifinal appearances, including one runner-up and two third-place finishes.</i> • Communications Lead, Experimental Class in Electronic Information Science
<i>Established and managed the class's official WeChat public account.</i> • Member, PKU AI Innovation and Entrepreneurship Club • Member, PKU Blockchain Association | <div>Oct 2022–Present</div> <div>Sep 2023–Present</div> <div>Jun 2024–Present</div> <div>Jun 2024–Present</div> |
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